DOMINO TOWERS OF HEIGHT 2 AND OF HEIGHT $n-1$, $n-2$, $n-3$: PROOF OF THE FOUR CONJECTURES IN OEIS A333650

MATH0

ABSTRACT. Let $T(n,k)$ be the number of domino towers with n bricks and k rows (OEIS A333650). We prove the four conjectures recorded in that entry by P. Kagey and P. Luschny in 2020: $T(n,2) = F_{n+2} - 1$ for $n \geq 2$, and $T(n,n-1)$, $T(n,n-2)$, $T(n,n-3)$ are the terms of index $n-2$, $n-3$, $n-4$ of the sequences A339252, A339254, A339029. For the near-diagonals, the row-by-row recursion on rows normalized up to translation closes on thirteen states, each counted by 2^h times an explicit polynomial of degree at most 3 in the number h of remaining rows. To our knowledge the near-diagonal identities are new (the height-two case is elementary and was in effect observed by Luschny); the entry still lists all four as conjectures. The theorem is verified in Lean 4 against a frozen, fidelity-reviewed statement that math0 formalized from the OEIS entry and checked against its data; it is not part of formal-conjectures.

1. INTRODUCTION

A *domino tower* is a stack of bricks of width 2 and height 1 in the plane, arranged in rows so that (1) each row is offset from the row below by half a brick, (2) the bottom row is contiguous, and (3) each brick above the bottom row is supported from below by at least half a brick (this is the definition in A264746 and A320314 [14], by Kagey). As in A333650, stability is not required, towers are counted up to translation, and mirror images are different. There are 3^{n-1} domino towers with n bricks; the count goes back to the theory of directed animals with a compact source [7, 1] (see Zeilberger [12] and Bóna's *Handbook* [2, pp. 25–27]). Towers of other pieces, and convex towers, are counted in [5, 3, 4, 8].

The OEIS entry A333650 [13], created by Peter Kagey on March 31, 2020, refines 3^{n-1} by height: $T(n,k)$ is the number of domino towers with n bricks and k rows. It lists rows 1–10 (rows up to 18 in a b-file), states $T(n,1) = 1$ and $T(n,n) = 2^{n-1}$, and records four conjectures (quoted in Section 2): $T(n,2) = \text{A000071}(n+2)$ (Kagey), $T(n,n-1) = \text{A339252}(n-2)$ (Kagey), $T(n,n-2) = \text{A339254}(n-3)$ and $T(n,n-3) = \text{A339029}(n-4)$ (Luschny). We prove all four (Theorem 2).

Method. For height two, $T(n,2) = \sum_b \binom{b+1}{n-b}$, a diagonal binomial sum. For a tower with n bricks and $n-e$ rows (e bricks beyond one per row), the recursion over rows normalized up to translation involves thirteen states when $e \leq 3$, each counted by 2^h times an explicit polynomial in the number h of remaining rows; the right-hand sequences have the same closed forms.

Status and prior work. All four statements are still marked "Conjecture" in A333650 (revision #68, August 12, 2022). The entries A339252, A339254, A339029 [15] define the right-hand sequences by recurrences or a generating function and record their closed forms (Corollary 11; for A339252 by K. Ryde, for A339254 by S. Spezia), but they do not connect them to towers. For height two, Luschny remarks in A339494 [16] (December 2020) that the towers of height at most 2, classified by base size, form "the binomial triangle $C(k, n-k)$ without the first column, triangle A030528", which "highlights the connection between the Pascal triangle and the

Date: October 19, 2026.

Fibonacci numbers"; a formula in A339495 [16] presupposes conjecture (1). This is essentially our argument for (1), but no proof is recorded. The math0 referees found no proof or counterexample of any of the four conjectures: the works on domino towers they read [7, 1, 12, 2, 5, 3, 4, 8] count towers by bricks, base size or convexity, not by height; Kagey's questions on Mathematics Stack Exchange concern symmetric towers and towers without the half-brick offset [9]; and web searches for the polynomials of Corollary 11 and for SeqFan discussions found nothing beyond the OEIS entries themselves. We therefore classify the result as *new*, the height-two case being elementary. A333650 is not among the OEIS statements of formal-conjectures [6]. Its remark that A038622 "appears to give" the counts by base size is not treated here.

2. Statement

The COMMENTS section of A333650 [13] contains, verbatim:

```
Conjecture:  For n > 1, T(n,2) = A000071(n+2).
Conjecture:  T(n,n-1) = A339252(n-2).  - _Peter Kagey_, Nov 21 2020
Conjecture:  T(n,n-2) = A339254(n-3).  - _Peter Luschny_, Nov 29 2020
Conjecture:  T(n,n-3) = A339029(n-4).  - _Peter Luschny_, Dec 01 2020
```

and Luschny's summary (December 1, 2020) $T(2n+k, n+k) = d_n(n+k-1)$ for $k \geq 1$, $0 \leq n \leq 3$, with explicit d_n (Corollary 11). The first conjecture is unsigned, hence by convention by the entry's author, Kagey (March 31, 2020). The entry's definition reads: "$T(n, k)$ gives the number of domino towers of height k consisting of n bricks. The towers must have a contiguous base of bricks, and each brick must be at least half supported below by another brick. The stacks do not need to be stable." The offset is 1, so $1 \leq k \leq n$.

Definition 1. Positions are measured in half bricks: a brick with left end $x \in \mathbb{Z}$ occupies $[x, x+2)$. Let $B_b = \{0, 2, \ldots, 2b-2\}$. A *tower of height k with n bricks* is a sequence $(L_0, \ldots, L_{k-1})$ of finite subsets of $\mathbb{Z}$ (the left ends of the bricks of row i, row 0 at the bottom) such that (a) every L_i is nonempty; (b) $L_0 = B_b$ for some $b \geq 1$; (c) $|x - y| \geq 2$ for distinct x, y in the same row; (d) $x \equiv i \pmod 2$ for $x \in L_i$; (e) for every $x \in L_{i+1}$ there is $y \in L_i$ with $|x - y| \leq 1$ (so the bricks overlap in length $2 - |x - y| \geq 1$); (f) $\sum_i |L_i| = n$. $T(n, k)$ is the number of towers of height k with n bricks.

Let F_m be the Fibonacci numbers ($F_0 = 0$, $F_1 = 1$), and let $a_1, a_2, a_3 \colon \mathbb{N} \to \mathbb{Z}$ be A339252, A339254, A339029 (offset 0):

$$a_1(0), a_1(1), a_1(2) = 1, 4, 11, \quad a_1(m+3) = 4a_1(m+2) - 4a_1(m+1);$$
$$a_2(0), \ldots, a_2(4) = 1, 7, 24, 70, 193, \quad a_2(m+5) = 6a_2(m+4) - 12a_2(m+3) + 8a_2(m+2);$$
$$\textstyle\sum_{m \geq 0} a_3(m) x^m = (1 + 4x - 20x^2 + 8x^3 + 33x^4 - 4x^5 - 33x^6)/(1-2x)^4.$$

Theorem 2. *(1) $T(n, 2) = F_{n+2} - 1$ for $n \geq 2$. (2) $T(n, n-1) = a_1(n-2)$ for $n \geq 2$. (3) $T(n, n-2) = a_2(n-3)$ for $n \geq 3$. (4) $T(n, n-3) = a_3(n-4)$ for $n \geq 4$.*

Lean formalization. The statement was formalized by math0 in `lean/Math0/Statements/OEIS/A333650.lean`, reviewed for fidelity and frozen (SHA-256 `20320a51...b96302`) before any proof attempt. Its definitions read (verbatim; the import, comments, docstrings, blank lines, and all lemmas and auxiliary definitions omitted):

```
namespace Math0.Statements.A333650
open Finset
def base (b : ℕ) : Finset ℤ := (range b).image fun j : ℕ => 2 * (j : ℤ)
def IsTower (n k : ℕ) (t : List (Finset ℤ)) : Prop :=
  t.length = k ∧ (∀ L ∈ t, L.Nonempty) ∧
  (∃ b, 0 < b ∧ t.head? = some (base b)) ∧
  (∀ L ∈ t, ∀ x ∈ L, ∀ y ∈ L, x ≠ y → 2 ≤ |x - y|) ∧
```

```
  (∀ (i : ℕ) (hi : i < t.length), ∀ x ∈ t[i], Even (x - i)) ∧
  (∀ (i : ℕ) (hi : i + 1 < t.length), ∀ x ∈ t[i + 1], ∃ y ∈ t[i], |x - y| ≤ 1) ∧
  (t.map Finset.card).sum = n
noncomputable def T (n k : ℕ) : ℕ := {t | IsTower n k t}.ncard
def A000071 (n : ℕ) : ℤ := (Nat.fib n : ℤ) - 1
def A339252 : ℕ → ℤ
  | 0 => 1
  | 1 => 4
  | 2 => 11
  | n + 3 => 4 * A339252 (n + 2) - 4 * A339252 (n + 1)
def A339254 : ℕ → ℤ
  | 0 => 1
  | 1 => 7
  | 2 => 24
  | 3 => 70
  | 4 => 193
  | n + 5 => 6 * A339254 (n + 4) - 12 * A339254 (n + 3) + 8 * A339254 (n + 2)
open PowerSeries in
noncomputable def A339029 (n : ℕ) : ℤ :=
  coeff n ((1 + 4 * X - 20 * X ^ 2 + 8 * X ^ 3 + 33 * X ^ 4 - 4 * X ^ 5 - 33 * X ^ 6 : ℤ⟦X⟧) *
    invOfUnit ((1 - 2 * X) ^ 4) 1)
def statement_col2 : Prop :=
  ∀ n : ℕ, 1 < n → (T n 2 : ℤ) = A000071 (n + 2)
def statement_diag1 : Prop :=
  ∀ n : ℕ, 2 ≤ n → (T n (n - 1) : ℤ) = A339252 (n - 2)
def statement_diag2 : Prop :=
  ∀ n : ℕ, 3 ≤ n → (T n (n - 2) : ℤ) = A339254 (n - 3)
def statement_diag3 : Prop :=
  ∀ n : ℕ, 4 ≤ n → (T n (n - 3) : ℤ) = A339029 (n - 4)
def statement : Prop :=
  statement_col2 ∧ statement_diag1 ∧ statement_diag2 ∧ statement_diag3
end Math0.Statements.A333650
```

`IsTower n k t` is Definition 1 for $t = [L_0, \dots, L_{k-1}]$, with (d) as `Even (x - i)`. The half-brick offset is forced by the data: if a brick could also sit exactly on another, then $T(2,2) = 3$, not the listed 2. `T_eq_Tc` proves that T equals a computable count `Tc` (the row recursion of Section 3.1), so the set of towers is finite and `ncard` is not the junk value 0; with `a_spec`, `a_data` checks all 55 listed terms (rows 1–10) by `decide +kernel`, i.e. by evaluation in the kernel. The right-hand sequences use their own OEIS definitions; `A339029_eq` identifies A339029 with a computable sequence `A339029rec` given by the recurrence read off from $(1-2x)^4$, and data lemmas check 40, 32, 30, 29 listed terms of A000071, A339252, A339254, A339029. Values are compared in $\mathbb{Z}$. Conjectures (2)–(4) state no range; they are read for every n at which the index on the right is a natural number. `statement_check` verifies the four identities for $n \le 10$ by `decide +kernel`. Luschny's summary is not formalized separately.

3. Proof

3.1. **The row recursion.** For finite $L \subseteq \mathbb{Z}$ let $\operatorname{supp} L = (L-1) \cup (L+1)$, the positions available to bricks resting on L. The statement file defines

$$\operatorname{ways}(L, 0, m) = [m = 0], \qquad \operatorname{ways}(L, h+1, m) = \sum_{j=1}^{m-h} \sum_{L' \subseteq \operatorname{supp} L,\ |L'|=j} \operatorname{ways}(L', h, m-j)$$

($m - h$ truncated at 0), and proves that $\operatorname{ways}(L, h, m)$ counts the sequences $(L_1, \dots, L_h)$ of nonempty sets with $L_1 \subseteq \operatorname{supp} L$, $L_{i+1} \subseteq \operatorname{supp} L_i$ and $\sum |L_i| = m$ (the bound $j \le m - h$ holds because the other h sets need h elements). In particular $\operatorname{ways}(L, h, m) = 0$ if $m < h$ (`ways_eq_zero`).

Lemma 3 (`T_eq_Tc`, statement file)**.** *For $k \geq 1$, $T(n,k) = \sum_{b=1}^{n} \mathrm{ways}(B_b, k-1, n-b)$.*

Proof. Given $L_0 = B_b$, (c)–(e) are equivalent to $L_{i+1} \subseteq \operatorname{supp} L_i$ for all i: if $x \in L_{i+1}$, $y \in L_i$ and $|x-y| \leq 1$, then $x - y$ is odd by (d), so $x = y \pm 1$. Conversely, the inclusions give (e), give (d) by induction from the even set B_b, and (d) gives (c). Finally $b = |B_b| \leq n$. □

3.2. **Height two.**

Proposition 4 (`col2`)**.** *For $n \geq 2$, $T(n,2) = F_{n+2} - 1$.*

Proof. For $r \geq 1$, only $j = r$ contributes to $\mathrm{ways}(L, 1, r)$, so $\mathrm{ways}(L,1,r) = \binom{|\operatorname{supp} L|}{r}$ (`ways_one`); and for $b \geq 1$, $\operatorname{supp} B_b = \{2i - 1 : 0 \leq i \leq b\}$ has $b+1$ elements (`card_supported_base`). For $n = m+2$, the term $b = n$ of Lemma 3 is $\mathrm{ways}(B_n, 1, 0) = 0$, so $T(m+2,2) = \sum_{b=1}^{m+1} \binom{b+1}{m+2-b} = \sum_{b=0}^{m} \binom{b+2}{m+1-b}$. By Mathlib's `Nat.fib_succ_eq_sum_choose`, $F_{m+4} = \sum_{i=0}^{m+3} \binom{i}{m+3-i}$. Here the terms $i = 0, 1$ vanish, the term $i = m+3$ is 1, and $i = b+2$ gives $\binom{b+2}{m+1-b}$. Hence $F_{m+4} = T(m+2,2) + 1$. □

3.3. **Translation invariance and excess.** For nonempty finite L let $\operatorname{nrm} L = L - \min L$, and put $V(S,e,h) = \mathrm{ways}(S, h, h+e)$, the number of ways to place h more rows with excess e on the row S.

Lemma 5 (`ways_map`, `ways_nrm`, `ways_succ`)**.** $\mathrm{ways}(L + c, h, m) = \mathrm{ways}(L, h, m)$ *for $c \in \mathbb{Z}$, hence* $\mathrm{ways}(\operatorname{nrm} L, h, m) = \mathrm{ways}(L, h, m)$*; and*

$$V(L, e, h+1) = \sum_{j=0}^{e} \sum_{L' \subseteq \operatorname{supp} L,\ |L'| = j+1} V(\operatorname{nrm} L', e-j, h).$$

Proof. Since $\operatorname{supp}(L + c) = \operatorname{supp}(L) + c$, the map $L' \mapsto L' + c$ is a bijection between the j-subsets of $\operatorname{supp} L$ and those of $\operatorname{supp}(L + c)$; invariance follows by induction on h. The recursion gives $\mathrm{ways}(L, h+1, h+1+e) = \sum_{i=1}^{e+1} \sum_{|L'|=i} \mathrm{ways}(L', h, h+1+e-i)$; put $i = j+1$ and use invariance. □

Lemma 6 (`Tc_diag`)**.** *For $n \geq e+1$ and $h = n - e - 1$, $T(n, n-e) = \sum_{b=0}^{e} V(B_{b+1}, e-b, h)$.*

Proof. By Lemma 3, $T(n, n-e) = \sum_{b=1}^{n} \mathrm{ways}(B_b, h, n-b)$. For $b \geq e+2$, $n - b < h$ and the term vanishes; for $b = b' + 1 \leq e+1$, $n - b = h + (e - b')$. □

3.4. **Thirteen states.** Let $\mathcal{S}$ be the set of the thirteen pairs (S, e) of Table 1, with the integer polynomials $p_{S,e}$ listed there: the bases B_b with excess $e \leq 4 - b$, and three gapped pairs that the recursion produces.

Lemma 7 (`closure_check`, `coef_check`, `base_check`)**.** *For every $(S,e) \in \mathcal{S}$: (i) $(\operatorname{nrm} L', e - j) \in \mathcal{S}$ for all $0 \leq j \leq e$ and $L' \subseteq \operatorname{supp} S$ with $|L'| = j+1$; (ii) $\sum_{j=0}^{e} \sum_{L'} p_{\operatorname{nrm} L', e-j}(x) = 2\, p_{S,e}(x+1)$ in $\mathbb{Z}[x]$, with L' as in (i); (iii) $128\, V(S,e,3) = 2^3 p_{S,e}(3)$.*

Proof. These are finite computations, summarized in Table 1. For example, from $(\{0,2\}, 1)$, with $\operatorname{supp}\{0,2\} = \{-1, 1, 3\}$, the singletons give $(\{0\}, 1)$ three times and the pairs give $(\{0,2\}, 0)$ twice and $(\{0,4\}, 0)$ once, so (ii) reads $2(128 + 144(x+1)) = 3(-32 + 96x) + 2 \cdot 192 + 256$. In Lean the polynomials are coefficient lists (`tab`), `nx` maps the coefficients of p to those of $2p(x+1)$ (`pe_nx`, by `ring`), and (ii) is checked on the coefficients of $x^0, \ldots, x^3$. Each of (i)–(iii) is one `decide +kernel`; for (iii) the kernel evaluates $\mathrm{ways}(S, 3, 3+e)$ by the recursion. □

Proposition 8 (`main`)**.** *For all $h \geq 3$ and $(S,e) \in \mathcal{S}$, $128\, V(S,e,h) = 2^h p_{S,e}(h)$.*

TABLE 1. The states (S, e); one step of Lemma 5 (the multiset of the $(\operatorname{nrm} L', e-j)$); the polynomial $p_{S,e}$ with $128\,V(S,e,h) = 2^h p_{S,e}(h)$ for $h \geq 3$; and $V(S,e,3)$.

(S,e)	one step	$p_{S,e}(h)$	$V(S,e,3)$
$(\{0\},0)$	$2(\{0\},0)$	128	8
$(\{0\},1)$	$2(\{0\},1)+(\{0,2\},0)$	$-32+96h$	16
$(\{0,2\},0)$	$3(\{0\},0)$	192	12
$(\{0\},2)$	$2(\{0\},2)+(\{0,2\},1)$	$-104+28h+36h^2$	19
$(\{0,2\},1)$	$3(\{0\},1)+2(\{0,2\},0)+(\{0,4\},0)$	$128+144h$	35
$(\{0,2,4\},0)$	$4(\{0\},0)$	256	16
$(\{0\},3)$	$2(\{0\},3)+(\{0,2\},2)$	$-228+h+30h^2+9h^3$	18
$(\{0,2\},2)$	$3(\{0\},2)+2(\{0,2\},1)+(\{0,4\},1)+(\{0,2,4\},0)$	$80+174h+54h^2$	68
$(\{0,2,4\},1)$	$4(\{0\},1)+3(\{0,2\},0)+2(\{0,4\},0)+(\{0,6\},0)$	$416+192h$	62
$(\{0,2,4,6\},0)$	$5(\{0\},0)$	320	20
$(\{0,4\},0)$	$4(\{0\},0)$	256	16
$(\{0,4\},1)$	$4(\{0\},1)+3(\{0,2\},0)+2(\{0,4\},0)+(\{0,6\},0)$	$416+192h$	62
$(\{0,6\},0)$	$4(\{0\},0)$	256	16

Proof. Induction on h, starting from Lemma 7(iii). By Lemma 5, the induction hypothesis (applicable by Lemma 7(i)) and Lemma 7(ii), $128\,V(S,e,h+1) = \sum_{j,L'} 128\,V(\operatorname{nrm} L', e-j, h) = 2^h \sum_{j,L'} p_{\operatorname{nrm} L', e-j}(h) = 2^{h+1} p_{S,e}(h+1)$. □

For $e \leq 3$ let $P_e = \sum_{b=0}^{e} p_{B_{b+1}, e-b}$; these pairs lie in $\mathcal{S}$ (`base_mem`), and by Table 1 (lemmas `R1`–`R3`; all by `decide +kernel`)

$$P_1 = 160+96h, \qquad P_2 = 280+172h+36h^2, \qquad P_3 = 588+367h+84h^2+9h^3.$$

Proposition 9 (`diag`). *For $e \leq 3$, $n \geq e+4$ and $h = n-e-1$, $128\,T(n,n-e) = 2^h P_e(h)$ (with $P_0 = 128$).*

This is Proposition 8 summed over Lemma 6.

3.5. **The right-hand sequences.**

Lemma 10 (`A339252_closed`, `A339254_closed`, `A339029_closed`). *For $e \in \{1,2,3\}$ and $h \geq e$, $128\,a_e(h) = 2^h P_e(h)$.*

Proof. By `A339029_eq`, $a_3(m+7) = 8a_3(m+6) - 24a_3(m+5) + 32a_3(m+4) - 16a_3(m+3)$ with $a_3(0), \ldots, a_3(6) = 1, 12, 52, 168, 497, 1412, 3879$. So for $h \geq e$, $a_e(h+e+1)$ is obtained from $a_e(h), \ldots, a_e(h+e)$ by the recurrence with characteristic polynomial $(x-2)^{e+1}$, which $2^h P_e(h)$ also satisfies, as $\deg P_e \leq e$. Induct on windows of $e+1$ consecutive values; the base window $h = e, \ldots, 2e$ is closed by `norm_num`, the steps by `linear_combination`. □

3.6. **Proof of Theorem 2.** Part (1) is Proposition 4. Let $e \in \{1,2,3\}$ and $n \geq e+1$. If $n \leq 10$, the identity is `statement_check`. If $n \geq 11$, then $n \geq e+4$ and $h = n-e-1 \geq 7$, and Proposition 9 and Lemma 10 give $128\,T(n,n-e) = 2^h P_e(h) = 128\,a_e(h)$ with $h = n-2, n-3, n-4$. □

Corollary 11. *$T(n,n-1) = 2^{n-4}(3n-1)$ for $n \geq 3$, $T(n,n-2) = 2^{n-8}(9n^2-11n+22)$ for $n \geq 5$, and $T(n,n-3) = 2^{n-11}(9n^3-24n^2+127n-112)$ for $n \geq 7$. Equivalently $T(n,n-e) = d_e(n-e-1)$ for $n \geq 2e+1$, where $d_1(m) = 2^{m-3}(10+6m)$, $d_2(m) = 2^{m-5}(70+43m+9m^2)$, $d_3(m) = 2^{m-7}(588+367m+84m^2+9m^3)$; this is Luschny's summary for $1 \leq n \leq 3$.*

This follows from Theorem 2 and Lemma 10, as $2^h P_e(h)/128 = d_e(h)$; Luschny's $T(2n+k, n+k)$ is $T(N, N-e)$ with $e = n$, $N = 2e+k$. The case d_0 is $T(n,n) = 2^{n-1}$. The corollary is not stated in Lean.

4. Machine verification

The proof is in `lean/Math0/Frontier/OEIS/A333650.lean` (394 lines) of the math0 repository, `https://github.com/Foundation-0/math0`. It imports the frozen statement file (unchanged, same SHA-256) and `lean/Math0/Audit.lean`. Its main declaration is `Math0.Frontier.A333650.result`, assembled from the private lemmas `col2`, `diag1`, `diag2`, `diag3` as in Section 3, and the file ends with `#math0_settles Math0.Frontier.A333650.result Math0.Statements.A333650.statement`. All finite checks use `decide +kernel` (evaluation by the kernel, which unlike `native_decide` adds no axiom). Neither file contains `sorry`, `native_decide`, `axiom` or `set_option`. The command `#math0_settles`, defined in `lean/Math0/Audit.lean` for statements that math0 formalizes itself, fails unless (1) `statement` is a universe-monomorphic `Prop`-valued definition; (2) `statement` and `result` use no axioms besides `propext`, `Classical.choice`, `Quot.sound` (so no `sorryAx` or `Lean.ofReduceBool`); (3) the type of `result` is syntactically the statement or its negation (up to metadata, no unfolding). It reports

```
math0 audit: 'Math0.Statements.A333650.statement' depends on axioms [Classical.choice, Quot.sound, propext]
math0 audit: 'Math0.Frontier.A333650.result' depends on axioms [Classical.choice, Quot.sound, propext]
math0 statement check: 'Math0.Frontier.A333650.result' proves 'Math0.Statements.A333650.statement'
```

We use Lean 4 [11] `v4.33.1` and Mathlib [10] `v4.33.1`, pinned in `lean/`; the repository also pins formal-conjectures [6] at commit `c77fdd4`, which this result does not use. To reproduce, run `scripts/verify.sh`: it builds all math0 results with `lake build`, re-running every audit, and checks that the pinned dependencies are unmodified. The single file is checked by `lake env lean Math0/Frontier/OEIS/A333650.lean` in `lean` (about a minute on our machine once Mathlib's compiled files are cached).

References

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